# Demo: Speaker

*ChatGPT edition — changing current becomes motion and sound*

Build, measure, and prove a paper-cup transducer

**Stop line and roles**

**Learner boundary:** isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

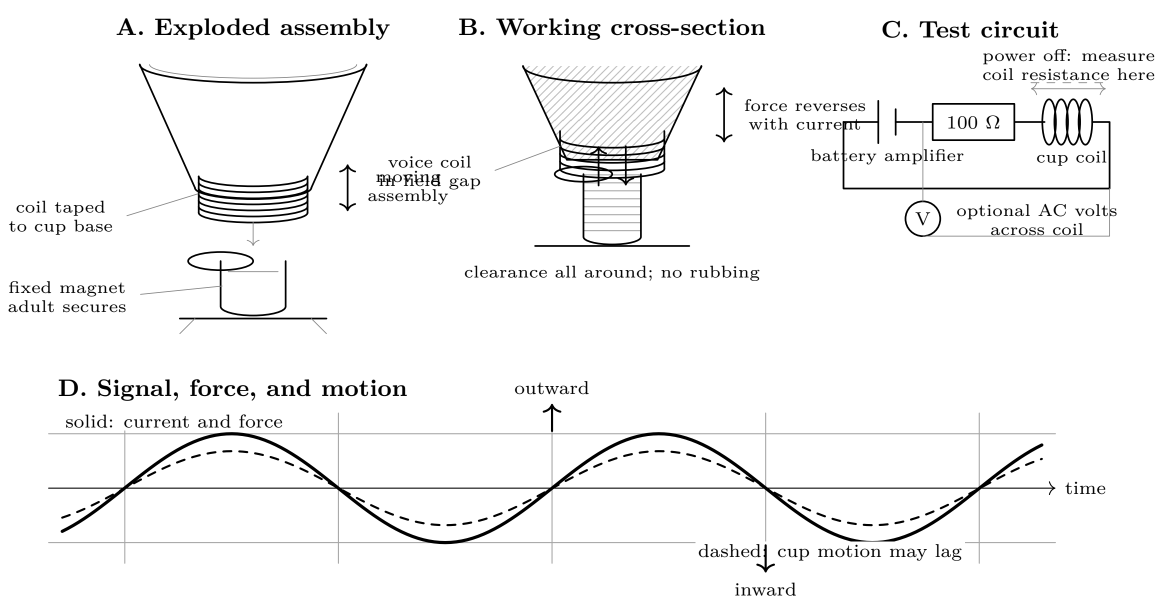
**What you need**

- Paper cup, card coil former, low-tack tape, paper flag
- 30–32 AWG insulated copper wire; sandpaper
- Ceramic magnet or adult-secured neodymium magnet
- Battery-powered, current-limited audio amplifier; tone source
- 100 Ω, 1 W series resistor; digital multimeter
- Ruler and optional phone spectrum display (observer only)

**Predict first**

Sketch what you expect for steady DC and for a slow alternating signal. Which part should move? Will doubling frequency double loudness? What measurement would distinguish “the coil is open” from “the cone is stuck”?

## Construction plate




## Investigate

### Stage 1: prove the coil before assembly

1. With all power disconnected, scrape both wire ends bright. Measure resistance directly across the coil. Record $R_{\text{coil}}$. An open display means repair the ends; near zero means look for a bypass or meter-lead error.
2. Gently flex each lead while watching the meter. The reading should stay steady. Ask: why can continuity pass while a

fragile joint still fails?

**Stage 2: assemble and inspect**

3. Tape the coil concentrically to the cup base. Adult secures the magnet. Center the coil over the magnet without contact. Move the cup lightly by hand: it must travel freely, silently, and return without scraping.
4. Adult checks that the 100 Ω series resistor is present, the amplifier is battery powered and at minimum level, and bare conductors cannot touch. Stop for heat, odor, damaged insulation, pain, or a loose magnet.

**Stage 3: first motion, then sound**

5. Apply a 20–30 Hz tone briefly at minimum level. Watch a paper flag on the cup. Increase only until motion is visible. Does the flag reverse at the same rate as the signal? Do not hold the moving coil.
6. Try 100, 250, 500, and 1000 Hz at the same source setting. Record whether sound is clear, faint, or distorted. If your meter reads AC volts reliably at that frequency, measure across the coil; otherwise mark it "meter bandwidth unknown" rather than treating the number as truth.

| Frequency | $V_{\text{coil}}$ | Observed level | Motion / quality |
|---|---|---|---|
| 20–30 Hz | ________ | ________ | ____________________ |
| 100 Hz | ________ | ________ | ____________________ |
| 250 Hz | ________ | ________ | ____________________ |
| 500 Hz | ________ | ________ | ____________________ |
| 1000 Hz | ________ | ________ | ____________________ |

**Troubleshoot from evidence**

| Evidence | Likely cause | Power-off check |
|---|---|---|
| Open resistance | Broken lead or enamel remains | Scrape ends; flex-test; find the discontinuity. |
| Stable resistance, no motion | No signal or bad connection | Verify amplifier on a known safe load; inspect every joint. |
| Motion but scraping | Coil off-center or magnet loose | Re-center; adult re-secures magnet. |
| Sound only at high level | Large gap, heavy cup, weak coupling | Reduce gap without contact; remove excess tape. |
| Buzz at one frequency | Loose tape or cup resonance | Press tape with power off; repeat just above and below that frequency. |
| Heat, odor, distortion | Excess drive or shorted turns | Disconnect at once; inspect and remeasure before deciding whether to retire the coil. |

**Final proof**

Reconnect after every power-off check. A successful build must pass all four:

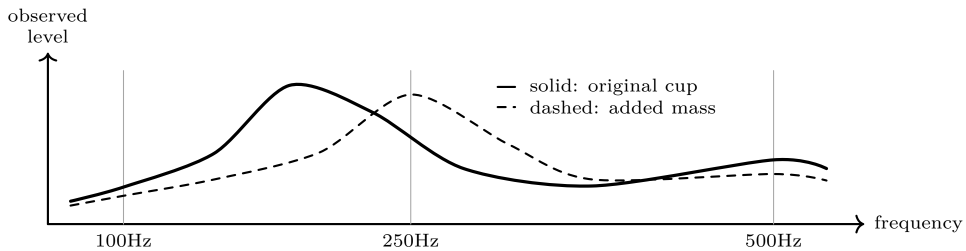
- Coil resistance is finite, repeatable, and unchanged by gentle lead flex.
- The unpowered cup moves freely without rubbing.
- A low-frequency signal produces visible alternating motion.
- At least two commanded audio frequencies produce distinguishable, repeatable sound, with no heat, odor, loose magnet, or unexpected contact.

## Explain from the evidence

The amplifier supplies changing current. In the permanent magnet's field, the coil experiences a force whose direction follows current direction. Because the coil is attached to the cup, the cup moves air. Steady DC can make one displacement and heating, but not a sustained tone. Loudness need not rise with frequency: coil impedance, cup mass and stiffness, magnetic gap, resonances, and the ear all shape the response. The measurements separate an electrical claim (a continuous coil receives a changing signal) from mechanical evidence (free, alternating motion) and acoustic evidence (repeatable sound).

## Change one thing

Add one removable paper clip near the cup rim, then repeat only 100, 250, and 500 Hz at the unchanged source setting. Predict first. Plot two simple response traces—frequency horizontally, observed level vertically—and circle any frequency where the added mass changes buzz, level, or motion. A phone spectrum display may confirm the strongest frequency, but it does not replace the resistance, motion, and stop-line checks.




| Frequency | Original level | With mass | What changed? |
| --- | --- | --- | --- |
| 100 Hz | ________ | ________ | ______________ |
| 250 Hz | ________ | ________ | ______________ |
| 500 Hz | ________ | ________ | ______________ |

**Questions.** Did added mass move the strongest response higher or lower? Which observation supports that claim? Was commanded frequency the same as the strongest measured peak? Which conclusion is justified by the evidence, and which remains only a plausible explanation?

| **Evidence record** | |
| --- | --- |
| Prediction | ____________________________ |
| One change | ____________________________ |
| Observation | ____________________________ |
| Claim + because | ____________________________ |

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.